과제 4

III.

(i) $U = \frac{X}{\|X\|}$, $R = \|X\|$. $X = RU$

$f(u,r) = f_X(ru)\,|J|\,\mathbb{1}\{r>0\}\,\mathbb{1}\{u \in S^{d-1}\}$

$= f_X(r)\,|J|\,\mathbb{1}\{r>0\}\,\mathbb{1}\{u \in S^{d-1}\}$

$= \underbrace{g(r)\,|J|\,\mathbb{1}\{r>0\}}_{=h_1(r)}\,\underbrace{\mathbb{1}\{u \in S^{d-1}\}}_{h_2(u)}$

$= h_1(r)\,h_2(u)$

$\Rightarrow R \perp\!\!\!\perp U \iff \frac{X}{\|X\|} \perp\!\!\!\perp \|X\|$

$h_2(u) = 1 \cdot \mathbb{1}\{u \in S^{d-1}\}$

which is pdf of $\mathrm{unif}(S^{d-1})$

$\Rightarrow \frac{X}{\|X\|} \sim \mathrm{Unif}(S^{d-1})$

$\boxtimes$

(ii)

$$U = \frac{X}{\|X\|} \rightsquigarrow U_j = \frac{X_j}{\|X\|}, \quad U_i U_j = \frac{X_i X_j}{\|X\|}$$

$$\text{Let } Q_j = \begin{pmatrix} 1 & & & & & 0 \\ & 1 & & & & \\ & & \ddots & & & \\ & & & -1 & & \\ & & & & \ddots & \\ 0 & & & & & 1 \\ & & & & & & 1 \end{pmatrix} \text{ ; orthogonal}$$

jth column

$$U \overset{d}{=} Q_j U$$

$$\mathbb{E} U_j = \mathbb{E}[Q_j U] = \mathbb{E}[-U_j] = -\mathbb{E} U_j$$

$$\Rightarrow \mathbb{E} U_j = \mathbb{E}\left[\frac{X_j}{\|X\|}\right] = 0$$

For $i \neq j$, $U \overset{d}{=} Q_i U$

$$\mathbb{E} U_i U_j = \mathbb{E}[(Q_i U)_i (Q_i U)_j]$$

$$= \mathbb{E}[(-U_i) U_j]$$

$$= -\mathbb{E}[U_i U_j]$$

$$\Rightarrow \mathbb{E} U_i U_j = 0 \quad \forall i \neq j$$

Let P_{ij} be permutation matrix

Sending ith coordinate to jth coordinate.

$\Rightarrow P_{ij}^{-1} = P_{ij}$ & $P_{ij}^T = P_{ij}$; orthogonal.

$\Rightarrow U \stackrel{d}{=} P_{ij} U$

$\mathbb{E} U_i^2 = \mathbb{E} U_j^2 =: a \quad \forall i, j$

$$1 \stackrel{\|U\|=1}{=} \mathbb{E}[\|U\|^2] = \mathbb{E}\left[\sum_{k=1}^{d} U_k^2\right]$$

$$= \sum_{k=1}^{d} \mathbb{E}[U_k^2]$$

$$= da$$

$$a = \frac{1}{d} = \mathbb{E} U_i^2$$

$$\Rightarrow \mathbb{E}\left[\frac{X_i X_j}{\|X\|^2}\right] = \begin{cases} \frac{1}{d} & , i=j \\ 0 & , i \neq j \end{cases}$$

$\boxtimes$

$\boxed{2}$.

(i) $r = \dfrac{D^{-1}(I-H)y}{\hat{\sigma}}$, $D = diag(\sqrt{1-h_{ii}})$

$\exists P$ s.t. $I-H = P(P^TP)^{-1}P^T$ & P is orthogonal
$= PP^T$

$\varepsilon^T(I-H)\varepsilon = \varepsilon^T PP^T\varepsilon$, $P^T\varepsilon \sim N(0, \sigma^2 P^TP) = N(0, \sigma^2 I)$

$\varepsilon \overset{d}{=} P^Te$, $\dfrac{e}{\|e\|} \perp\!\!\!\perp \|e\|$

$\dfrac{P^Te}{\|P^Te\|} \perp\!\!\!\perp \|P^Te\| \iff \dfrac{P^Te}{\sqrt{e^TPP^Te}} \perp\!\!\!\perp \sqrt{e^TPP^Te}$

$\mathbb{E}\left[\dfrac{D^{-1}PP^Te}{\sqrt{\frac{e^TPP^Te}{n-p-1}}}\right] = D^{-1}P\,\mathbb{E}\left[\dfrac{P^Te}{\sqrt{\frac{e^TPP^Te}{n-p-1}}}\right]$

$$\underset{0}{\underset{\|}{\mathbb{E}[P^T\varepsilon]}} = \mathbb{E}\left[\dfrac{P^Te}{\sqrt{\frac{\varepsilon^TPP^T\varepsilon}{n-p-1}}}\right] \cdot \underbrace{\mathbb{E}\left[\sqrt{\dfrac{\varepsilon^TPP^T\varepsilon}{n-p-1}}\right]}_{>0}$$

$(\because \dfrac{\varepsilon^TPP^T\varepsilon}{\sigma^2} \sim \chi^2(n-p-1))$

$\Rightarrow \mathbb{E}\left[\dfrac{P^Te}{\sqrt{\frac{\varepsilon^TPP^Te}{n-p-1}}}\right] = 0$

$\Rightarrow \mathbb{E}[r] = 0$

$\Rightarrow \mathbb{E}[r_i] = 0$

$$r_i^2 = \frac{e_i^2}{\hat{\sigma}^2(1-h_{ii})} \Rightarrow \mathbb{E}\, r_i^2 = \frac{n-p-1}{1-h_{ii}} \mathbb{E}\left[\frac{e_i^2}{SSE}\right]$$

$$e_i = u_i(I-H)y \quad , \quad u_i = (0, 0, \dots, \overset{i\text{th}}{1}, 0, 0)$$

$$\frac{e_i^2}{SSE} = \frac{y^T(I-H)u_i^T u_i (I-H)y}{y^T(I-H)y}$$

$$= \frac{\epsilon^T(I-H)u_i^T u_i (I-H)\epsilon}{\epsilon^T(I-H)\epsilon}$$

$$= \frac{\epsilon^T PP^T u_i^T u_i PP^T\epsilon}{\epsilon^T PP^T\epsilon} \quad , \quad W := \frac{P^T\epsilon}{\|P^T\epsilon\|} \sim N(0, \sigma^2 I)$$

$$= \frac{W^T P^T u_i^T u_i P W}{W^T W}$$

$$\frac{P^T\epsilon}{\|P^T\epsilon\|} \perp\!\!\!\perp \|P^T\epsilon\|$$

$$\Rightarrow \frac{u_i P W}{W^T W} \perp\!\!\!\perp W^T W = \epsilon^T PP^T \epsilon$$

$$\Rightarrow \mathbb{E}\left[\frac{W^T P^T u_i^T u_i P W}{W^T W}\right] \times \mathbb{E}[W^T W] = \mathbb{E}[W^T P^T u_i^T u_i P W]$$

$$= tr(P^T u_i^T u_i P \sigma^2 I) + 0$$

$$= (1-h_{ii})\sigma^2, \quad PP^T = I-H$$

$$\Rightarrow \mathbb{E}\left[\frac{w^T P^T u_i u_i P w / \sigma^2}{w^T w / \sigma^2}\right] \times \underbrace{\mathbb{E}\left[\frac{w^T w}{\sigma^2}\right]}_{\substack{\shortparallel \\ n-p-1}} = 1 - h_{ii}$$

$$\Rightarrow \mathbb{E}\left[\frac{w^T P^T u_i^T u_i P w / \sigma^2}{w^T w / \sigma^2}\right] = \mathbb{E}\left[\frac{e_i^2}{SSE}\right] = \frac{1-h_{ii}}{n-p-1}$$

$$\Rightarrow \mathbb{E}[r_i^2] = \frac{n-p-1}{1-h_{ii}} \times \frac{1-h_{ii}}{n-p-1} = 1$$

$$\therefore \mathbb{E} r_i = 0 \ , \ var(r_i) = 1 \qquad \blacksquare$$

(ii) $cov(r_i, r_j) \overset{var(r_i)=1}{=} Corr(r_i, r_j)$

$$= \mathbb{E}[r_i r_j] - \mathbb{E} r_i \mathbb{E} r_j$$

$$= \mathbb{E}[r_i r_j]$$

$$= \mathbb{E}\left[\frac{e_i e_j}{\hat{\sigma}^2 \sqrt{1-h_{ii}}\sqrt{1-h_{jj}}}\right], \ e_i = u_i(I-H)y$$

$$= \mathbb{E}\left[\frac{y^T(I-H) u_i^T u_j (I-H) y}{\frac{y^T(I-H)y}{n-p-1} \sqrt{1-h_{ii}}\sqrt{1-h_{jj}}}\right]$$

$$= \mathbb{E}\left[\frac{\epsilon^T P P^T u_i^T u_j P P^T \epsilon}{\epsilon^T P P^T \epsilon}\right] \frac{n-p-1}{\sqrt{1-h_{ii}}\sqrt{1-h_{jj}}}$$

$$\& \ w = \frac{P^T e}{\|P^T e\|}$$

$$= \mathbb{E}\left[\frac{w^T P^T U_i U_j P w}{w^T w}\right] \frac{n-p-1}{\sqrt{1-h_{ii}}\sqrt{1-h_{jj}}}$$

$$\Rightarrow \mathbb{E}\left[\frac{w^T P^T U_i U_j P w / \sigma^2}{w^T w / \sigma^2}\right] \times \mathbb{E}[w^T w]$$

$$= \mathbb{E}[w^T P U_i U_j P w / \sigma^2]$$

$\because \frac{w}{\|w\|} \perp\!\!\!\perp \|w\|$

$\Rightarrow \frac{U_i P w}{\|w\|} \perp\!\!\!\perp \|w\|$

$$= tr(P^T U_i U_j P) + 0$$

$$= tr(U_i (I-H) U_j)$$

$$= \begin{cases} -h_{ij} & , i \neq j \\ 1-h_{ii} & , i = j \end{cases}$$

$cov(r_i, r_j)$

$$= \mathbb{E}\left[\frac{w^T P^T U_i U_j P w / \sigma^2}{w^T w / \sigma^2}\right] \frac{n-p-1}{\sqrt{1-h_{ii}}\sqrt{1-h_{jj}}}, \quad \mathbb{E}[w^T w] = n-p-1$$

$$= \begin{cases} \frac{-h_{ij}}{n-p-1} \times \frac{n-p-1}{\sqrt{1-h_{ii}}\sqrt{1-h_{jj}}} & , i \neq j \\ \frac{1-h_{ii}}{n-p-1} \times \frac{n-p-1}{1-h_{ii}} & , i = j \end{cases}$$

$$= \begin{cases} \frac{-h_{ij}}{\sqrt{1-h_{ii}}\sqrt{1-h_{jj}}} & , i \neq j \\ 1 & , i = j \end{cases}$$

$\blacksquare$

$\boxed{3}$

(i) Let $X^* = (X \quad y)$

$$H^* = X^*(X^{*T}X^*)^{-1}X^{*T}$$

$$= (X \quad y)\begin{pmatrix} X^TX & X^Ty \\ y^TX & y^Ty \end{pmatrix}^{-1}\begin{pmatrix} X^T \\ y^T \end{pmatrix}$$

$$= (X \quad y)\begin{pmatrix} (X^TX)^{-1} + (X^TX)^{-1}X^Ty(y^T(I-H_x)y)^{-1}y^TX(X^TX)^{-1} & -(X^TX)^{-1}X^Ty(y^T(I-H_x)y)^{-1} \\ -(y^T(I-H_x)y)^{-1}y^TX(X^TX)^{-1} & (y^T(I-H_x)y)^{-1} \end{pmatrix}$$

$$\begin{pmatrix} X^T \\ y^T \end{pmatrix}$$

$$= H_x + H_xy(y^T(I-H_x)y)^{-1}y^TH_x - y(y^T(I-H_x)y)^{-1}y^TH_x$$
$$- H_xy(y^T(I-H_x)y)^{-1}y^T + y(y^T(I-H_x)y)^{-1}y^T$$

Here, $y^T(I-H_x)y = e^Te$

$$= H_x + \frac{1}{e^Te}(H_xyy^TH_x - yy^TH_x - H_xyy^T + yy^T)$$

$$= H_x + \frac{1}{e^Te}[(yy^T - H_xyy^T)(I-H_x)]$$

$$= H_x + \frac{1}{e^Te}[(I-H_x)yy^T(I-H)]$$

$$= Hx + \frac{ee^T}{e^Te} \Rightarrow h_{ii}^* = h_{ii} + \frac{e_i^2}{e^Te}$$

Since H^* is hat matrix, $\frac{1}{n} \le h_{ii}^* \le 1$

$$\Rightarrow \frac{1}{n} \le h_{ii} + \frac{e_i^2}{e^Te} \le 1$$

$\boxtimes$

(II) $$DFFITS_i = \frac{\hat{y}_i - \hat{y}_{i(i)}}{\sqrt{\hat{\sigma}_{(i)}^2 h_{ii}}}$$

$$= \frac{h_{ii} e_i}{(1-h_{ii}) \hat{\sigma}_{(i)} \sqrt{h_{ii}}}$$

$$\frac{\hat{\sigma}\sqrt{p+1}}{\hat{\sigma}_{(i)}} \sqrt{D_i} = \frac{\hat{\sigma}\sqrt{p+1}}{\hat{\sigma}_{(i)}} \sqrt{\frac{r_i^2}{p+1} \frac{h_{ii}}{1-h_{ii}}}$$

$$= \frac{\hat{\sigma}}{\hat{\sigma}_{(i)}} \sqrt{\frac{h_{ii}}{1-h_{ii}}} \, r_i$$

$$= \frac{\hat{\sigma}}{\hat{\sigma}_{(i)}} \sqrt{\frac{h_{ii}}{1-h_{ii}}} \frac{e_i}{\hat{\sigma}\sqrt{1-h_{ii}}}$$

$$= \frac{e_i}{\hat{\sigma}_{(i)}} \frac{\sqrt{h_{ii}}}{1-h_{ii}}$$

$$= \frac{\hat{y}_i - \hat{y}_{i(i)}}{\sqrt{\hat{\sigma}_{(i)}^2 h_{ii}}} = DFFITS_i$$

$\boxtimes$

四.

(1) $\frac{\partial \|y - X\beta - \theta d_i\|^2}{\partial \theta}$

$= -2d_i^T(y - X\beta - \theta d_i) = 0$

$\Rightarrow d_i^T y - d_i^T X\beta - \theta d_i^T d_i = 0$

$\Rightarrow y_i - x_i^T \beta - \theta = 0$

$\Rightarrow \hat{\theta} = y_i - x_i^T \hat{\beta}$

$\frac{\partial \|y - X\beta - \theta d_i\|^2}{\partial \beta} = -2X^T(y - X\beta - \theta d_i) = 0$

$\Rightarrow X^T X \beta = X^T y - x_i^T \theta$

$\Rightarrow \tilde{\beta} = (X^T X)^{-1} X^T y - (X^T X)^{-1} x_i^T \theta$

$= \hat{\beta} - (X^T X)^{-1} x_i^T \theta$

$\hat{\theta} = y_i - x_i^T \tilde{\beta}$

$= y_i - x_i^T \hat{\beta} + h_{ii} \hat{\theta}$

$\Rightarrow (1 - h_{ii}) \hat{\theta} = e_i$

$\Rightarrow \hat{\theta} = \frac{e_i}{1 - h_{ii}}$

$$e_i = y - x_i^T \hat{\beta}$$
$$= (d_i^T - x_i^T (X^TX)^{-1} X^T) y \quad ; \text{ linear comb of normal.}$$

$$\mathbb{E}[e_i] = \mathbb{E}[(d_i^T - x_i^T (X^TX)^{-1} X^T) y]$$
$$= (d_i^T - x_i^T (X^TX)^{-1} X^T)(X\beta + \theta d_i)$$
$$= x_i^T \beta - x_i^T \beta + \theta - h_{ii}\theta$$
$$= (1 - h_{ii})\theta$$

$$\mathbb{E}[\hat{\theta}] = \theta$$

$$\mathrm{var}(e_i) = \sigma^2 (d_i^T - x_i^T (X^TX)^{-1} X^T)(d_i - X (X^TX)^{-1} x_i)$$
$$= \sigma^2 (1 - h_{ii} - h_{ii} + h_{ii})$$
$$= \sigma^2 (1 - h_{ii})$$

$$\mathrm{var}(\hat{\theta}) = \frac{\sigma^2}{1 - h_{ii}}$$

$$\therefore \hat{\theta} \sim N\left(\theta, \frac{\sigma^2}{1 - h_{ii}}\right) \quad \boxed{\text{//}}$$

(ii) $y = X\beta + \theta d_i + e$

$= X^* \beta^* + e, \quad X^* = (X \;\; d_i), \quad \beta^* = \begin{pmatrix} \beta \\ \theta \end{pmatrix}$

$H_0 : \theta = 0$

$$F = \frac{y^T (H_{X^*} - H_X) y / 1}{y^T (I - H_{X^*}) y / n-p-2}$$

Here, $(I - H_X) d_i =: X_{d|t}$

$$H_{X^*} - H_X = H_{X_{d|t}}$$

$$= X_{d|t} (X_{d|t}^T X_{d|t})^{-1} X_{d|t}^T$$

$$= (I - H_X) d_i (d_i^T (I - H_X) d_i)^{-1} d_i^T (I - H_X)$$

$$= \frac{1}{1 - h_{ii}} (I - H_X) d_i d_i^T (I - H_X)$$

$$= \frac{1}{1 - h_{ii}} (I - H_X) K_i (I - H_X)$$

$$\Rightarrow y^T (H_{X^*} - H_X) y = \frac{y^T (I - H_X) K_i (I - H_X) y}{1 - h_{ii}} = \frac{e_i^2}{1 - h_{ii}}$$

$$y^T (I - H_{X^*}) y = y^T (I - H_X - H_{X_{d|t}}) y$$

$$= y^T \left(I - H_X - \frac{1}{1 - h_{ii}} (I - H_X) K_i (I - H_X)\right) y$$

$$= y^T (I - H_X - L) y$$

$$= (n - p - 2) \widehat{\sigma_{(i)}^2}$$

$$\frac{e_i}{1-h_{ii}} \sim N\left(\theta, \frac{\sigma^2}{1-h_{ii}}\right)$$

$$\frac{e_i}{\sqrt{1-h_{ii}}\,\sigma} \sim N\left(\frac{\theta}{\sigma}\sqrt{1-h_{ii}}, 1\right)$$

$$F = \frac{\frac{e_i^2}{1-h_{ii}} / 1}{(n-p-2)\,\hat{\sigma}^2_{(i)} / n-p-2} = \frac{e_i^2}{\hat{\sigma}^2_{(i)}(1-h_{ii})}$$

$$t_i = \frac{e_i}{\hat{\sigma}_{(i)}\sqrt{1-h_{ii}}} = \frac{\frac{e_i}{1-h_{ii}} / \sigma}{\sqrt{\frac{(n-p-2)\,\hat{\sigma}^2_{(i)}}{\sigma^2} / n-p-2}}$$

$$= \frac{N\left(\frac{\theta}{\sigma}\sqrt{1-h_{ii}}, 1\right)}{\sqrt{\chi^2(n-p-2) / n-p-2}}$$

$$\sim t\left(n-p-2, \frac{\theta}{\sigma}\sqrt{1-h_{ii}}\right)$$

$$(I-H_{X^*})(H_{X^*}-H_X)$$
$$= H_{X^*} - H_{X^*} - H_X + H_X$$
$$= 0 \;;\; N \perp \chi^2$$

$$t_2 \overset{H_0}{\sim} t(n-p-2, 0) = t(n-p-2)$$

$$\overset{H_1}{\sim} t\left(n-p-2, \frac{\theta}{\sigma}\sqrt{1-h_{22}}\right) \qquad \blacksquare$$

$\boxed{5}$.

$$\widehat{Err_m} = \overline{err} + \hat{\omega}$$; indirect estimation.

$$L(y_i, \hat{y}_i) = -2\log N(y_i \mid \hat{y}_i, \sigma_\varepsilon^2)$$

$$= -2\log \frac{1}{\sqrt{2\pi\sigma_\varepsilon^2}} e^{-\frac{(y_i-\hat{y}_i)^2}{2\sigma_\varepsilon^2}}$$

$$= \log 2\pi\sigma_\varepsilon^2 + \frac{(y_i-\hat{y}_i)^2}{\sigma_\varepsilon^2}$$

$$\overline{err} = \frac{1}{N}\sum_{i=1}^{N} L(y_i, \hat{y}_i) = \log 2\pi\sigma_\varepsilon^2 + \frac{\sum (y_i-\hat{y}_i)^2}{N\sigma_\varepsilon^2}$$

$$= \log 2\pi\sigma_\varepsilon^2 + \frac{SSE}{N\sigma_\varepsilon^2}$$

$$\omega = \mathbb{E}\left[Err_m - \overline{err}\right]$$

$$= \mathbb{E}\left[\frac{1}{N}\sum_{i=1}^{N}\mathbb{E}_{y_i^{new}}\left[L(y_i^{new}, \hat{y}_i) \mid \mathcal{T}\right] - \frac{1}{N}\sum_{i=1}^{N} L(y_i, \hat{y}_i)\right]$$

$$= \mathbb{E}\left[\frac{1}{N}\sum_{i=1}^{N}\mathbb{E}\left[\log 2\pi\sigma_\varepsilon^2 + \frac{(y_i^{new}-\hat{y}_i)^2}{\sigma_\varepsilon^2} \mid \mathcal{T}\right] - \frac{1}{N}\sum_{i=1}^{N}\left[\log 2\pi\sigma_\varepsilon^2 - \frac{(y_i-\hat{y}_i)^2}{\sigma_\varepsilon^2}\right]\right]$$

$$= \mathbb{E}\Big[\frac{1}{N\sigma_\varepsilon^2}\sum_{i=1}^{N} \mathbb{E}\big[(y_i^{new}-\hat{y}_i)^2 \mid S\big] - \frac{(y_i-\hat{y}_i)^2}{N\sigma_\varepsilon^2}\Big]$$

$$= \frac{1}{\sigma_\varepsilon^2}\,\mathbb{E}\Big[\underbrace{\frac{1}{N}\sum_{i=1}^{N}\mathbb{E}\big[(y_i^{new}-\hat{y}_i)^2 \mid S\big]}_{(*)} - \underbrace{\frac{1}{N}\sum_{i=1}^{N}(y_i-\hat{y}_i)^2}_{(**)}\Big]$$

$$(*) = \frac{1}{N}\sum_{i=1}^{N}\mathbb{E}\big[(y_i^{new} - f(x_i) + f(x_i) - \mathbb{E}\hat{y}_i + \mathbb{E}\hat{y}_i - \hat{y}_i)^2 \mid S\big]$$

$$= \frac{1}{N}\sum_{i=1}^{N}\Big(\sigma_\varepsilon^2 + (y_i - \mathbb{E}\hat{y}_i)^2 + (f(x_i) - \mathbb{E}\hat{y}_i)^2 - 2(\hat{y}_i - \mathbb{E}\hat{y}_i)(f(x_i) - \mathbb{E}\hat{y}_i)\Big)$$

$$\Rightarrow \mathbb{E}(*) = \frac{1}{N}\sum_{i=1}^{N}\Big(\sigma_\varepsilon^2 + \operatorname{var}(\hat{y}_i) + (f(x_i) - \mathbb{E}\hat{y}_i)^2\Big)$$

$$(**) := \frac{1}{N}\sum_{i=1}^{N}(y_i - f(x_i) + f(x_i) - \mathbb{E}\hat{y}_i + \mathbb{E}\hat{y}_i - \hat{y}_i)^2$$

$$\mathbb{E}(**) = \frac{1}{N}\sum_{i=1}^{N}\Big(\sigma_\varepsilon^2 + \operatorname{var}(\hat{y}_i) + (f(x_i) - \mathbb{E}\hat{y}_i)^2 - 2\operatorname{cov}(y_i, \hat{y}_i)\Big)$$

$$\Rightarrow \frac{1}{\sigma_\varepsilon^2}\big(\mathbb{E}(*) - \mathbb{E}(**)\big)$$

$$= \frac{1}{\sigma_\varepsilon^2 N}\sum_{i=1}^{N} 2\operatorname{cov}(y_i, \hat{y}_i)$$

$$= \frac{2}{\sigma_\varepsilon^2 N}\operatorname{tr}\big(\operatorname{cov}(y, Hy)\big)$$

$$= \frac{2(p+1)}{N}$$

$$\widehat{\mathrm{Err}}_{in} = \log 2\pi\sigma_\varepsilon^2 + \frac{SSE}{N\sigma_\varepsilon^2} + \frac{2(p+1)}{N}$$

minimizing $\widehat{\mathrm{Err}}_{in}$

$\iff$ minimizing $\frac{SSE}{N\sigma_\varepsilon^2} + \frac{2(p+1)}{N}$

$\iff$ minimizing $SSE + 2(p+1)\sigma_\varepsilon^2$

Here, $C_p = \frac{SSE}{\sigma_\varepsilon^2} + 2(p+1) - N$

minimizing C_p

$\iff$ minimizing $\frac{SSE}{\sigma_\varepsilon^2} + 2(p+1) - N$

$\iff$ minimizing $\frac{SSE}{\sigma_\varepsilon^2} + 2(p+1)$

$\iff$ minimizing $SSE + 2(p+1)\sigma_\varepsilon^2$

$\iff$ minimizing $\widehat{\mathrm{Err}}_{in}$

$\blacksquare$

$\boxed{6}$.

$$\mathbb{E}[N \times \mathrm{Err}_{in}]$$

$$= \mathbb{E}\Big[\sum_{i=1}^{N} \mathbb{E}_{y_i^{new}}\big[(y_i^{new} - x_i\hat{\beta})^2 \mid \mathcal{J}\big]\Big]$$

$$= \mathbb{E}\Big[\sum_{i=1}^{N} \mathbb{E}\big[(x_i\beta - x_i\hat{\beta} + e^{new})^2 \mid \mathcal{J}\big]\Big]$$

$$= \mathbb{E}\Big[\sum_{i=1}^{N}\big(\mathbb{E}[(x_i\beta - x_i\hat{\beta})^2] + \sigma_\varepsilon^2\big)\Big]$$

$$= \mathbb{E}\Big[\sum_{i=1}^{N}\big(\mathbb{E}[(u_i^T f(x) - u_i H(f(x)+e))^2] + \sigma_\varepsilon^2\big)\Big]$$

$$= \mathbb{E}\Big[\sum_{i=1}^{N} \mathbb{E}[(\underbrace{u_i^T(I-H)f(x)}_{=\eta_i} - u_i^T H e)^2 + \sigma_\varepsilon^2]\Big]$$

$$= \mathbb{E}\Big[\sum_{i=1}^{N}\big(\mathbb{E}(\eta_i - u_i^T H e)^2 + \sigma_\varepsilon^2\big)\Big]$$

$$= \mathbb{E}\Big[\sum_{i=1}^{N}\big(\mathbb{E}[\eta_i^2 - 2\eta_i u_i^T H e + e^T H u_i u_i^T H e] + \sigma_\varepsilon^2\big)\Big]$$

$$= \mathbb{E}\Big[\sum_{i=1}^{N}\big(\eta_i^2 + \mathrm{tr}(u_i^T H u_i \sigma^2) + 0 + \sigma_\varepsilon^2\big)\Big]$$

$$= \sum_{i=1}^{N}\big(\eta_i^2 + h_{ii}\sigma_\varepsilon^2 + \sigma_\varepsilon^2\big)$$

$$= \sum_{i=1}^{N}\big(\eta_i^2 + \sigma^2(1+h_{ii})\big)$$

$$\mathrm{LOCV} = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{e_i}{1-h_{ii}}\right)^2 = \frac{1}{N}\sum_{i=1}^{N}\frac{e_i^2}{(1-h_{ii})^2}$$

$$e_i = u_i^T(I-H)y = u_i^T(I-H)(X\beta+\epsilon)$$

$$= \underbrace{u_i^T(I-H)X\beta}_{\eta_i} + u_i^T(I-H)\epsilon$$

$$= \eta_i + u_i^T(I-H)\epsilon$$

$$\mathbb{E}[e_i^2] = \mathbb{E}[\eta_i^2 + \epsilon^T(I-H)K_i(I-H)\epsilon$$
$$+ 2\eta_i^T u_i^T(I-H)\epsilon]$$

$$\& \quad \mathbb{E}[\eta_i^T u_i^T(I-H)\epsilon] = 0$$

$$= \mathbb{E}[\eta_i^2 + \epsilon^T(I-H)K_i(I-H)\epsilon]$$

$$= \eta_i^2 + \mathbb{E}[\epsilon^T(I-H)K_i(I-H)\epsilon]$$

$$= \eta_i^2 + \mathrm{tr}(u_i^T(I-H)u_i\sigma^2 I) + 0$$

$$= \eta_i^2 + \sigma^2(1-h_{ii})$$

$$\mathbb{E}[N \times \mathrm{LOCV}] = \sum_{i=1}^{N}\frac{\eta_i^2 + \sigma^2(1-h_{ii})}{(1-h_{ii})^2}$$

$$E[N \times LOOV] - E[N \times Err_{in}]$$

$$= \sum_{i=1}^{N} \frac{\mu_i^2 + \sigma_\varepsilon^2 (1-h_{ii})}{(1-h_{ii})^2} - \sum_{i=1}^{N} \left(\mu_i^2 + \sigma_\varepsilon^2 (1+h_{ii})\right)$$

$$= \sum_{i=1}^{N} \left(\frac{\mu_i^2}{(1-h_{ii})^2} - \mu_i^2 \right) + \sum_{i=1}^{N} \left(\frac{\sigma_\varepsilon^2}{1-h_{ii}} - \frac{\sigma_\varepsilon^2 (1-h_{ii}^2)}{1-h_{ii}} \right)$$

$$= \sum_{i=1}^{N} \frac{1-(1-h_{ii})^2}{(1-h_{ii})^2} \mu_i^2 + \sum_{i=1}^{N} \frac{h_{ii}^2}{1-h_{ii}} \sigma_\varepsilon^2$$

$$= \sum_{i=1}^{N} \frac{h_{ii}(2-h_{ii})}{(1-h_{ii})^2} \mu_i^2 + \sum_{i=1}^{N} \frac{h_{ii}^2}{1-h_{ii}} \sigma_\varepsilon^2$$

$\blacksquare$